Igor Bispo de Moraes Coelho Correia

AI Specialist | Computer Vision & Deep Learning Engineer

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igorbispo99.github.io

PROFESSIONAL SUMMARY

AI Specialist with 5+ years applying deep learning and computer vision to mission-critical domains — road infrastructure, SAR remote sensing, medical imaging, and automotive safety. Proven track record across the full ML lifecycle: architecture design, training, optimization, and production deployment. MSc in Electrical Engineering (Signal Processing) from the University of Brasília.

TECHNICAL STACK

Deep Learning & CV	PyTorch, TensorFlow, OpenCV, YOLO, Faster R-CNN, U-Net, Transformers, HuggingFace
Languages	Python, C++, C#
Image Processing	SAR imagery, medical imaging (CT/panoramic), embedded video, depth estimation, visual enhancement
MLOps & Infra	Docker, Kubernetes, Apache Airflow, Nagios, Linux on-premise, AWS, GCP
Data	PostgreSQL, SQL Server, Elasticsearch
Tools	CVAT (annotation), Django, Git

WORK EXPERIENCE

Senior AI Specialist
2025 – present

FPFtech · Manaus, Brazil (hybrid)

Developed computer vision systems for automotive safety: object detection and classification in compliance with international industry regulations, depth estimation, and visual enhancement. Worked across the full CV pipeline — preprocessing, image quality assessment, deep learning inference, and performance monitoring on resource-constrained embedded hardware.

Postgraduate Lecturer — Machine Learning & Deep Learning
2025 – present

IPOS · Remote

Teaching Machine Learning with Python and Deep Learning with Python in the AI for Civil Construction specialization program. Responsible for curriculum design, course delivery, and assessment.

AI Analyst
2022 – 2025

National Transportation Confederation (CNT) · Brasília, Brazil

Managed the full AI model lifecycle — design, training, and deployment — for the CNT Road Survey, Brazil's largest highway assessment program. Built and maintained detection, classification, and analysis systems covering 40+ road variables (pavement defects, signage, viaducts, guardrails). Stack: Python, C++, C#, TensorFlow, PyTorch, OpenCV, PostgreSQL, SQL Server, Apache Airflow.

Postgraduate Researcher
2021 – 2025

University of Brasília (UnB) · Brasília, Brazil

Curupira project: built a deforestation and wildfire detection system for Sentinel-1 SAR imagery using PyTorch and deep convolutional networks. Gigasistêmica project: designed and trained models for osteoporosis and atheroma detection in CT and dental panoramic images (U-Net, semantic segmentation). Supervised annotation pipelines in CVAT.

Data Scientist
2021 – 2022

Zerum IT Engineering · Brasília, Brazil

Natural language analysis and fiscal text classification for fraud and anomaly detection. Developed LSTM and Transformer models for time-series data. Production classification API with Django. Stack: Python, TensorFlow, HuggingFace, Elasticsearch.

EDUCATION

MSc in Electrical Engineering — Signal Processing

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University of Brasília (UnB) *2022 – 2024*

BSc in Computer Science
University of Brasília (UnB) *2017 – 2021*

PUBLICATIONS

Correia, I. B. M. C. et al. *A Comprehensive Assessment of Optimized YOLO Architectures for Edge and Real-Time Applications*. In: 2026 IEEE 2nd International Conference on Consumer Technology (ICCT-Pacific). DOI: 10.1109/ICCT-Pacific69083.2026.11518725.

Correia, I. B. M. C. et al. *Importance-Aware Quantization Strategy for Layer-Sensitive YOLO Architectures*. In: 2026 IEEE 2nd International Conference on Consumer Technology (ICCT-Pacific). DOI: 10.1109/ICCT-Pacific69083.2026.11518766.

Correia, I. B. M. C. et al. *Detection and Segmentation of Carotid Atheroma Calcification in Dental Panoramic Radiographs Using a Hybrid Deep Learning Model*. Preprint. Available at: researchsquare.com/article/rs-8078056/v1.

Correia, I. B. M. C.; Farias, M. et al. *Comparative Study of Deforestation Detection in Sentinel-1 Scenes of the Amazon Forest*. In: Proceedings of the XLI Brazilian Symposium on Telecommunications and Signal Processing (SBrT2023). DOI: 10.14209/sbrt.2023.1570923825.

Correia, I. B. M. C. *Deforestation Detection in SAR Images using Deep Neural Networks*. MSc Dissertation — University of Brasília, Brasília, 2024. Available at: repositorio.unb.br/handle/10482/50921.

LANGUAGES

Portuguese	Native	**English**	Advanced	**Spanish**	Intermediate